GUIDELINES

2019 IUSTI-Europe guideline for the management of anogenital warts

Abstract

This guideline is an update of the 2011 European Guideline for the Management of Anogenital Warts. It aims to support best practice in the care of patients with anogenital warts by providing evidence-based recommendations on diagnosis, treatment, follow-up, and patient advice. It is intended for use by healthcare professionals in sexual health or dermato-venereology clinics in Europe, though it may be adapted for other settings where anogenital warts are managed. As a European guideline, recommendations should be adapted according to national circumstances and healthcare systems.

Despite the availability of vaccines preventing HPV types 6 and 11—responsible for >95% of anogenital warts—these lesions remain a common and significant health issue. A previous systematic review of randomized controlled trials (RCTs) on anogenital wart treatments was updated. Key updates in this guideline include:

- Updated background on the prevalence, natural history, and transmission of HPV infection and anogenital warts
- Grading of key recommendations based on the strength of recommendation and quality of evidence
- Evaluation and inclusion of 5-fluorouracil, local interferon, and photodynamic therapy as potential second-line treatment options
- Updated evidence on the impact of HPV vaccination on anogenital wart incidence

Introduction and methodology

This document updates the 2011 European Guideline for the Management of Anogenital Warts. It provides evidence-based guidance on diagnosis, treatment, follow-up, and patient counseling to support optimal clinical care. The target audience includes healthcare professionals working in sexual health or dermatology clinics across Europe, although the guideline may be adapted for use in other settings.

Recommendations should be tailored to national contexts and healthcare systems. Clinicians and service providers are encouraged to develop local treatment algorithms that reflect their specific resources, treatment availability, and patient population needs.

The treatment recommendations from the 2011 guideline were based on a systematic review of RCTs. This review has been updated using the search strategy outlined in Table 1 to identify studies published since 2009. The literature search was conducted on 13 May 2019 using MEDLINE and EMBASE (2009–present).

Table 1. Search strategy

Combines with AND	Population (OR)	Intervention (OR)
	Keywords: genital wart, *anogenital wart*, anogenital-wart, anal wart, *condyloma*, venereal wart Plus appropriate MeSH or mapped headings	Keywords: podophyllotoxin, podofilox, podophyllin, cryotherapy, liquid nitrogen, imiquimod, aldara, TCA, trichloroacetic acid, BCA, bichloroacetic acid, curettage, excision, surgical removal, laser, electrosurgery, electrocautery, hyfrecation, ablati, *diathermy, 5-fluorouracil, polyphenon, photodynamic,* tretin* (to capture acitretin, tretinoin, isotretinoin), retinoids Plus appropriate MeSH or mapped headings

Existing systematic reviews on anogenital wart treatments were also considered. The quality of evidence and strength of recommendations were assessed using the grading system from the British HIV Association Guidelines Group.[1] National guidelines from the United Kingdom, Germany, and the United States were also reviewed.[2–4]

The full protocol for developing this guideline is available on the IUSTI website: https://iusti.org/wp-content/uploads/2020/04/ProtocolForProduction2020.pdf

Aetiology and transmission

Anogenital warts are benign epithelial proliferations occurring on the genitalia, anus, perianal area, and occasionally the inguinal or pubic regions. They are caused primarily by infection with human papillomavirus (HPV), most commonly low-risk types 6 and 11, which account for over 95% of cases.

HPV is transmitted through direct skin-to-skin or mucosal contact, typically during sexual activity. Transmission can occur even in the absence of visible warts, due to subclinical or latent infection. The incubation period varies widely, ranging from weeks to months, and spontaneous regression may occur. Factors associated with increased risk include young age, multiple sexual partners, and immunosuppression (e.g., HIV infection).